MILITARY SPECIFICATION

CLINOMETER, ELEVATION ML-591/U

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers the design and construction of one type of Clinometer ML 591/U, for shipboard use only.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-A-250/8A	Aluminum Alloy 5052, Plate and Sheet
QQ-A-325	Aluminum Alloy Bars, Rods, Wire and Special Shapes, Rolled Drawn, or Cold Finished, 6061
QQ-B-626	Brass, Leaded and Non-leaded, Rods, Shapes, Forgings and Flat Products with Finished Edges, Bars, Flat Wire and Strips
QQ-C-320	Chromium Plating
QQ-N-290	Nickel Plating
QQ-S-561	Solder, Silver
QQ-S-571	Solder, Lead Alloy, Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Form
QQ-S-763	Steel Bars, Shapes and Forgings, Corrosion Resisting

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WW-P-351	Pipe, Brass, Seamless, Iron-Pipe-Size, Standard
WW-P-460	Pipe-Fittings, Brass or Bronze, Screwed
WW-T-789	Tube, Aluminum Alloy, Round, Square, Rectangular, and other Shapes, Drawn, 6061 and 6062

Military

MIL-B-5687	Bearings, Sleeve, Washers, Thrust, Sintered, Metal Powder, Oil-impregnated
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8623	Adhesive, Epoxy, Resin, Metal to Metal Structural Bonding
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-P-15035	Plastic Sheet, Laminated, Thermosetting, Cotton-Fabric-Base, Phenolic-Resin
MIL-B-16540	Bronze, Phosphor, Castings
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Federal

FED-STD-NO. 151	Metals; Test Methods

Military

MIL-STD-NO. 129	Marking for Shipment and Storage

DRAWINGS

Bureau of Naval Weapons

62A142	Clinometer, Cloud Height ML-591/U

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(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

National Bureau of Standards

Handbook H-28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402)

3. REQUIREMENTS

3.1 Material - All parts shall be in accordance with Bureau of Naval Weapons drawing 62A142 and the applicable specification listed below:

a. Aluminum ..... FED QQ-A-250/8A
QQ-A-325
WW-T-789

b. Brass ..... FED QQ-B-626
WW-P-351
WW-P-460

c. Bronze MIL-B-16540

d. Oilite MIL-B-5687

e. Plastic MIL-P-15035

f. Solder ..... FED QQ-S-561
QQ-S-571

g. Steel, corrosion resistant FED QQ-S-763

3.1.1 Screws - All screws, nuts, and similar parts used in the assembly of the instrument shall conform to the Handbook H 28.

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3.2 Design and Construction - The design and construction shall be in accordance with Bureau of Naval Weapons drawing 62A142, except that part 7, on Figure 1, shall be nickel plated.

3.3 Surface Treatment - Surface treatment shall be in accordance with Bureau of Naval Weapons drawing 62A142 and the following specifications:

a. Anodic coating MIL-A-8625

b. Chromium plating ..... FED QQ-C-320

c. Nickel plating ..... FED QQ-N-290

3.4 Performance - (See Figure No. 1) The line of sight through the eyepiece and the crosshairs shall be parallel with the top of the sighting tube within one-half degree. The sighting tube (part No. 1) shall swivel easily without binding or scraping when the knurled hand knob (part No. 3) is loosened. When the knurled hand knob is tightened hand tight, the sighting tube shall be imobile. No more than one-quarter turn of the knurled hand knob shall be required to free or secure the sighting tube. With the knurled hand knob loosened, swiveling the sighting tube shall not cause the knob to turn. The index plate (part No. 2) shall swivel freely with the knurled thumbscrew (part No. 5) loosened and shall be held imobile when it is tightened finger tight. The upper assembly shall slide freely within the mounting bracket when the locking nut (part No. 7) is loose. When the instrument is secured in an extended position, with the locking nut tightened hand tight, it shall be capable of supporting at least twenty-five pounds.

3.5 Preproduction sample - One preproduction sample of the complete equipment covered by this specification shall be submitted for approval. The sample shall be made and assembled by tools and methods which are the same as those which will be used in quantity production of the equipment on contract.

3.6 Nameplate - Nameplates shall be in accordance with specification MIL-N-18307 and secured to the sighting tube by adhesive MIL-A-8623.

3.7 Workmanship - The clinometer shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, making of parts, welding and brazing, plating and freedom of parts from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspections specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction Model - In addition to the tests of paragraph 4.3, the preproduction model, if required under the terms of the contract or purchase order, shall be subjected to the salt spray test of Federal Standard No. 151.

4.3 Production Models - Each production model shall undergo the following inspection and tests:

4.3.1 Materials - In accordance with the applicable specification and Bureau of Naval Weapons drawing 62A142.

4.3.2 Design and Construction - In accordance with Bureau of Naval Weapons drawing 62A142.

4.3.3 Surface Inspection - Visual and in accordance with Bureau of Naval Weapons drawing 62A142.

4.3.4 Performance Inspection - In accordance with paragraph 3.4.

5. PREPARATION FOR DELIVERY

5.1 Marking for shipment - Shipping containers shall be marked in accordance with MIL-STD-129.

5.2 Preservation, Packaging and Packing - The Elevation Clinometer shall be preserved, packaged and packed in accordance with the requirements of Specification MIL-P-7936.

6. NOTES

6.1 Intended use - The clinometer is used for measuring the angle of elevation of a spot of light thrown on the base of a cloud by a ceiling light projector located at the opposite end of a measured baseline. This information is then used in computing ceiling height (cloud height).

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6.2 Ordering data - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Preproduction sample required.

c. Stock number of Clinometer.

d. Requirement for salt spray test.

e. Special packing if required.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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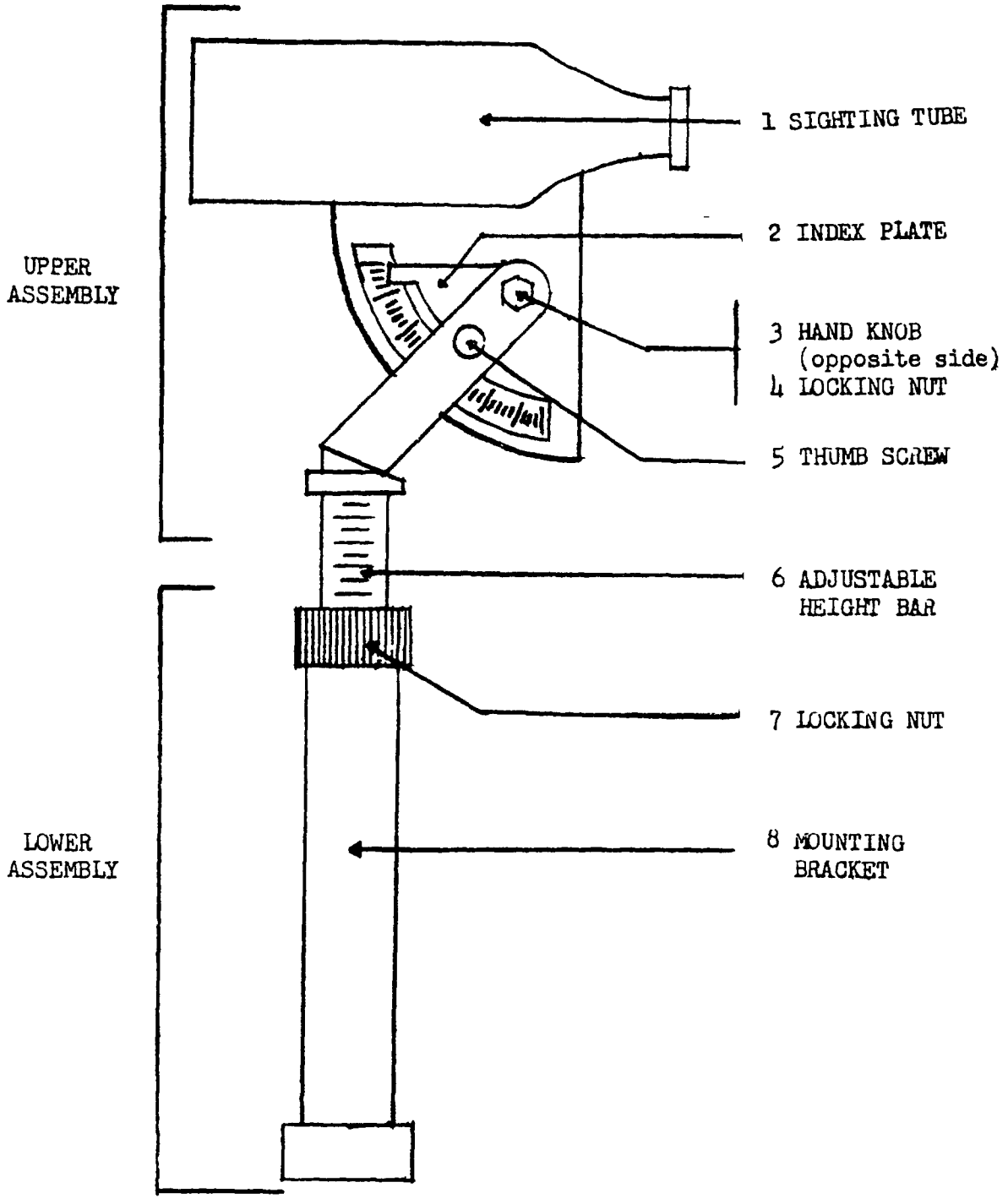


FIGURE 1. CLINOMETER, ELEVATION

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (6-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R009

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1 HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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